CS420 - MCQs

CS420 · Total: 100 questions · Nexora

Q1 **According to the course material, what is the primary difference between the content strategy of a traditional website and a mobile website?** EASY

- A **Traditional websites prioritize depth and completeness, while mobile websites focus only on essential [Correct]**
- B Traditional websites focus on speedy access to specific info, while mobile websites prioritize secondary information.
- C Traditional websites are built using native app wrappers, while mobile websites are built using Bootstrap.
- D Traditional websites require touchscreen navigation, while mobile websites rely solely on mouse clicks.

Correct Answer: A - Explanation:
The correct answer is A because the text states that traditional websites prioritize depth and completeness, whereas mobile users seek essential information and speedy access. Option B is incorrect because it reverses these priorities, falsely claiming traditional sites focus on speed and mobile on secondary info. Option C is incorrect because Bootstrap and PhoneGap are frameworks, not content strategies. Option D is incorrect because touchscreen navigation is a hardware/interaction difference, not a content strategy difference.

Q2 **In Chrome DevTools, what is the effect of holding the 'Shift' key and clicking on the color picker?** MEDIUM

- A It automatically sets the background color to blanched almond.
- B **It allows you to cycle through different color schemes such as RGB and HSL.** [Correct]
- C It opens the Network, Timeline, and Profiles tabs simultaneously.
- D It resets the border width and font size of the selected element to default.

Correct Answer: B - Explanation:
The correct answer is B because the course material explicitly states that 'Shift + clicking the color picker, allows you to cycle through the different color schemes. RGB HSL, and so on.' Option A is incorrect because 'blanched almond' is simply an example of a color the instructor manually typed, not an automatic result of shift-clicking. Option C is incorrect because the Network, Timeline, and Profiles tabs are separate panels in DevTools and are not opened by the color picker. Option D is incorrect because shift-clicking does not reset layout properties like border width or font size.

Q3 **To enable Developer Mode on many Android devices for mobile web debugging, what specific action must a developer perform?** EASY

- A Connect the device to a TV and cast a video in full-screen mode.
- B Open Chrome Canary and enter the IP address 127.0.0.1.
- C **Go to the device's settings, click on 'About Device', and click on 'Build Number' seven times.** [Correct]
- D Enable Port Forwarding on Port 9999 using Python's simple HTTP server.

Correct Answer: C - Explanation:
The correct answer is C because the text states that to turn on Developer Mode, you must go to settings, click 'About Device', and click 'Build Number' seven times. Option A is incorrect because casting video to a TV is a feature of Chrome screencasting, not a method to enable Developer Mode. Option B is incorrect because Chrome Canary is a desktop browser and cannot enable settings on an Android device. Option D is incorrect because setting up Python's HTTP server is part of reverse port forwarding, not device configuration.

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Q4 When setting up your development environment as described in the course, which browser versions are recommended to be installed on your laptop and mobile device respectively? MEDIUM

A Chrome Beta on the laptop and Chrome Canary on the mobile device.
B Chrome Canary on both the laptop and the mobile device.
C Chrome Beta on both the laptop and the mobile device.
D Chrome Canary on the laptop and Chrome Beta on the mobile device. [Correct]

Correct Answer: D - Explanation:
The correct answer is D because the text states: 'On my laptop, I have Chrome Canary and on my mobile device, I have Chrome Beta installed.' Option A is incorrect because it reverses the two browser versions between the laptop and mobile device. Options B and C are incorrect because they suggest installing the same browser version on both devices, which does not match the recommended setup.

Q5 Suppose you run Python's simple HTTP server on your local development machine on Port 9999. To access this local server on your mobile device over USB, what configuration must you set up in the Chrome Inspect page? HARD

A Enable Port Forwarding and add a rule mapping port 9999 to localhost (127.0.0.1:9999). [Correct]
B Set the viewport meta tag to width=9999 and initial-scale=1.
C Disable USB debugging and connect both devices to the same TV casting network.
D Use the iOS web kit debug proxy to forward Port 8080 to Port 9999.

Correct Answer: A - Explanation:
The correct answer is A because the text explains that to access a local server running on Port 9999 on a mobile device, you must set up Port Forwarding on the Chrome Inspect page with a rule for port 9999 on localhost (127.0.0.1). Option B is incorrect because viewport meta tags control layout scaling, not network routing. Option C is incorrect because disabling USB debugging would prevent the mobile device from communicating with the development machine. Option D is incorrect because the iOS web kit debug proxy is used for mobile Safari, not Chrome over USB.

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Q6 Who outlined the 'Mobile First' design approach in 2009? EASY

A Apple Developer Group
B Luke Wroblewski [Correct]
C The Google Chrome DevTools Team
D Python Software Foundation

Correct Answer: B - Explanation:
The correct answer is B because the text explicitly credits Luke Wroblewski with outlining the 'Mobile First' design approach in 2009. Option A is incorrect because Apple is credited with introducing the viewport meta tag, not the Mobile First philosophy. Option C is incorrect because the Chrome DevTools team created the debugging tools, not the design philosophy. Option D is incorrect because the Python Software Foundation is responsible for the Python language, which is unrelated to Mobile First.

Q7 The 'Mobile First' design approach is also known by which of the following terms? MEDIUM

A Reverse Port Forwarding
B Fluid Grid Layout
C **Progressive enhancement** [Correct]
D Adaptive Viewport Scaling

Correct Answer: C - Explanation:
The correct answer is C because the text states: 'This design approach is also known as "progressive enhancement."' Option A is incorrect because reverse port forwarding is a networking technique used to access local servers during debugging. Option B is incorrect because fluid grids are a component of responsive design, not another name for Mobile First. Option D is incorrect because adaptive viewport scaling is a browser rendering behavior, not a design philosophy.

Q8 When mobile browsers first emerged, they 'lied' about their window width to fit desktop content. If an original iPhone had a physical screen width of 320 pixels, what default width did its browser act as if it had? HARD

A 640 pixels
B 1024 pixels
C 768 pixels
D **980 pixels** [Correct]

Correct Answer: D - Explanation:
The correct answer is D because the text states that mobile browsers made the window act as if it were 980 pixels wide to fit desktop content on a 320-pixel screen. Option A is incorrect because 640 pixels is the physical resolution of some later iPhones, not the default lied width. Options B and C are incorrect because 1024 and 768 represent standard desktop screen dimensions, not the specific 980-pixel default width used by mobile browsers.

Q9 Which company originally provided the viewport meta tag to HTML to control the default screen scaling on mobile devices? MEDIUM

A **Apple** [Correct]
B Google
C Microsoft
D Mozilla

Correct Answer: A - Explanation:
The correct answer is A because the text states: 'In order to control this, Apple provided a viewport meta tag to be added to your HTML to control the default.' Option B is incorrect because Google developed Chrome and its DevTools, but did not originate the viewport tag. Options C and D are incorrect because Microsoft and Mozilla adopted the tag later but were not its creators according to the course material.

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Q10 When making live changes to viewport settings in your source code, why is it highly recommended to close the tab on your mobile device and reopen it, rather than just hitting reload? HARD

A Reloading the page will permanently disable USB debugging on the Android device.

B The Python HTTP server will crash if a viewport change is detected during a reload.

C The browser maintains the zoom level across page refreshes, which can prevent new viewport settings from [Correct]

D iOS devices will automatically lock the screen orientation to portrait if you reload.

Correct Answer: C - Explanation:
The correct answer is C because the text explains that the browser maintains the zoom level across page refreshes, meaning viewport changes in source code may not show up unless the tab is closed first. Option A is incorrect because reloading a page has no effect on the USB debugging connection state. Option B is incorrect because the Python server only serves files and does not crash based on browser viewport reloads. Option D is incorrect because reloading does not lock screen orientation on iOS.

Q11 What is defined as the 'user's visible area of a web page' which varies with the device? EASY

A The Canvas

B The FlexBox

C The Screen Cast

D The Viewport [Correct]

Correct Answer: D - Explanation:
The correct answer is D because the text defines the viewport as 'the user's visible area of a web page' which varies with the device. Option A is incorrect because the canvas refers to a fixed-size visual design layout surface. Option B is incorrect because FlexBox is a CSS layout tool used to stack elements. Option C is incorrect because Screen Cast is a DevTools feature that mirrors the mobile screen on a desktop.

Q12 Which of the following is NOT a downside or side effect of using a fixed-width viewport on mobile devices? MEDIUM

A The text and UI will automatically scale up and look extremely large on desktop screens. [Correct]

B The content is almost always going to be scaled and will not match the native resolution.

C Users of some devices will have to zoom to read text or interact with the UI.

D The UI elements (like dials or buttons) can become awfully small and hard to use on small screens.

Correct Answer: A - Explanation:
The correct answer is A because the text states that non-mobile desktop browsers do not respond to the viewport tag at all, meaning text and UI will not change on desktop. Option B is incorrect because scaling content away from native resolution is a documented downside of fixed-width viewports. Option C is incorrect because forcing users to zoom to read text is a major accessibility drawback of fixed widths. Option D is incorrect because small, hard-to-use UI elements are a direct consequence of scaling down a fixed width.

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Q13 **If you set the viewport meta tag to '<meta name="viewport" content="width=device-width">' on iOS, what happens when a user rotates the device from portrait to landscape mode if NO initial-scale is set?** HARD

A. The viewport width updates to 480 pixels and reflows the text at native size.
B. **The iOS web engine keeps the portrait viewport width and stretches/rescales the layout to fit the landscape** [Correct]
C. The page automatically reloads and prompts the user to allow USB debugging.
D. The font size is automatically reduced by 50% to prevent text overflow.

Correct Answer: B - Explanation:
The correct answer is B because the text states that on iOS, if you only set width to device-width without initial-scale, rotating the device keeps the portrait viewport width and stretches it across the landscape screen. Option A is incorrect because the viewport width only updates to the landscape width (e.g., 480px) if initial-scale=1 is set. Option C is incorrect because rotation does not trigger page reloads or USB debugging prompts. Option D is incorrect because iOS does not automatically shrink fonts by 50% upon rotation.

Q14 **To prevent iOS from incorrectly scaling text and layout when rotating from portrait to landscape, what is the recommended default boilerplate viewport meta-element?** MEDIUM

A. <meta name="viewport" content="width=980, initial-scale=0.5">
B. <meta name="viewport" content="width=device-width, user-scalable=no">
C. **<meta name="viewport" content="width=device-width, initial-scale=1">** [Correct]
D. <meta name="viewport" content="width=916, minimum-scale=1, maximum-scale=1">

Correct Answer: C - Explanation:
The correct answer is C because the text recommends using 'width=device-width, initial-scale=1' as the default boilerplate to ensure iOS scales correctly when flipping orientations. Option A is incorrect because it uses a fixed width of 980 and an incorrect scale. Option B is incorrect because disabling user scaling is bad for accessibility and not part of the standard boilerplate. Option D is incorrect because it uses a fixed width of 916 and restricts scaling.

Q15 **On Android, if a page with 'width=device-width' forcibly overflows due to an element being too wide, how does the browser behave during orientation changes, and how can a developer temporarily bypass this during testing?** HARD

A. Android crashes; the developer must use the iOS web kit debug proxy.
B. Android ignores the overflow; the developer must set user-scalable=no.
C. Android scales the page correctly; the developer does not need to take any action.
D. **Android tries to rescale and zooms the wrong way; the developer can set minimum-scale and maximum-scale to 1** [Correct]

Correct Answer: D - Explanation:
The correct answer is D because the text states that Android tries to rescale overflowing pages during orientation changes but gets it wrong, and setting minimum-scale and maximum-scale to 1 is the only way to prevent this. Option A is incorrect because Android does not crash, and the iOS proxy is irrelevant to Android. Option B is incorrect because user-scalable=no is not the recommended fix for this specific Android overflow bug. Option C is incorrect because Android does not scale it correctly by default.

Q16 **Why does the author advise against setting 'user-scalable=no' in production mobile web pages?** MEDIUM

- A **It is bad for accessibility because it prevents users from zooming.** [Correct]
- B It causes Chrome Canary to crash when screen casting is active.
- C It forces the viewport width to default back to 980 pixels.
- D It disables the Python simple HTTP server's port forwarding rules.

Correct Answer: A - Explanation:
The correct answer is A because the text states that disabling user zooming is bad for accessibility and annoys users. Option B is incorrect because user-scalable=no does not cause Chrome Canary or screen casting to crash. Option C is incorrect because disabling user scaling does not reset the viewport width to 980 pixels. Option D is incorrect because viewport properties have no impact on Python server configurations or port forwarding.

Q17 **If you want to size a button using CSS viewport units so that it always takes up exactly one-third of the width of the viewport, which CSS rule should you apply?** HARD

- A width: 33.33vh;
- B **width: 33.33vw;** [Correct]
- C width: 33.33vmin;
- D width: 33.33vmax;

Correct Answer: B - Explanation:
The correct answer is B because 'vw' represents a percentage of the viewport's width, so '33.33vw' corresponds to exactly one-third of the width. Option A is incorrect because 'vh' stands for viewport height, which would size the button relative to the screen's height instead of width. Options C and D are incorrect because 'vmin' and 'vmax' size elements based on the minimum or maximum of the viewport's dimensions, which would not guarantee exactly one-third of the width across different orientations.

Q18 **What do the CSS units 'vmin' and 'vmax' represent?** MEDIUM

- A The minimum and maximum zoom levels allowed by the viewport meta tag.
- B The minimum and maximum font sizes allowed on mobile Safari.
- C **Sizing based on the smallest or largest units of viewport width and height.** [Correct]
- D Sizing limits for FlexBox containers in portrait mode.

Correct Answer: C - Explanation:
The correct answer is C because the text states that 'vmin' and 'vmax' let you size things based on the smallest or largest units of width and height. Option A is incorrect because minimum and maximum zoom levels are controlled by 'minimum-scale' and 'maximum-scale' in the viewport tag. Option B is incorrect because font size limits are not defined by these CSS units. Option D is incorrect because FlexBox does not use these units as container limits.

Q19 **Which of the following is one of the three simple rules outlined in the course if a developer decides to offer a mobile-only site?** EASY

A Block access to the web application with a full-page ad for the native app.

B Disable zooming entirely by setting user-scalable to no.

C Ensure the mobile site has a fixed viewport width of exactly 980 pixels.

D **Put a canonical URL in the meta-information to keep the mobile and desktop sites together.** [Correct]

Correct Answer: D - Explanation:
The correct answer is D because the text explicitly lists putting a canonical URL in the meta-information as one of the three rules for mobile-only sites. Option A is incorrect because the text states developers should NOT block access with full-page ads. Option B is incorrect because disabling zoom is discouraged. Option C is incorrect because fixed viewports are historical and not a rule for mobile-only sites.

Q20 **How does the course distinguish between 'Fluid' and 'Responsive' websites?** MEDIUM

A **Fluid websites use percentages for widths to scale columns relative to one another, while Responsive** [Correct]

B Fluid websites use media queries for specific device sizes, while Responsive websites use only fixed pixel widths.

C Fluid websites are built using native app wrappers, while Responsive websites are built using Python's simple HTTP server.

D Fluid websites only work on desktop monitors, while Responsive websites only work on mobile phones.

Correct Answer: A - Explanation:
The correct answer is A because the text defines Fluid websites as using percentages for widths to scale columns relatively, and Responsive websites as using a fluid grid combined with media queries. Option B is incorrect because it reverses the definitions of Fluid and Responsive. Option C is incorrect because native app wrappers and Python servers are development tools, not website design types. Option D is incorrect because both fluid and responsive designs are meant to work across multiple device types.

Q21 **According to the source material, which of the following is the most commonly used media feature in media queries?** EASY

A **width** [Correct]

B orientation

C resolution

D aspect-ratio

Correct Answer: A - Explanation:
The source explicitly states that 'The most commonly used media feature is width.' Options B, C, and D are available media query expressions listed in the slides, but they are not identified as the most commonly used. Therefore, width is the correct answer because it is the primary feature used to target different screen sizes.

Q22 **Consider the following CSS code block:** MEDIUM

```
@media screen and (min-width: 480px) {
  #leftsidebar { width: 200px; float: left; }
  #main { margin-left: 216px; }
}
```

What is the behavior of the '#leftsidebar' element when the viewport width is 400 pixels?

A. It will float to the left of the page with a width of 200px.
B. **It will not float to the left and will instead be positioned on top of the content.** **[Correct]**
C. It will scale down to 100% width and maintain a margin-left of 216px.
D. It will trigger a horizontal scrollbar because the viewport is smaller than 480px.

Correct Answer: B - Explanation:
The media query '@media screen and (min-width: 480px)' only applies its block of CSS rules when the viewport width is 480 pixels or wider. Since the viewport in this scenario is 400 pixels, the rules inside the query are ignored. According to the source, if the viewport is less than 480 pixels, the menu (represented by #leftsidebar) will be on top of the content.

Q23 **Which CSS3 Flexbox property is used to modify the behavior of the 'flex-wrap' property by aligning flex lines instead of individual flex items?** MEDIUM

A. align-items
B. justify-content
C. **align-content** **[Correct]**
D. flex-flow

Correct Answer: C - Explanation:
The source defines 'align-content' as the property that 'Modifies the behavior of the flex-wrap property. It is similar to align-items, but instead of aligning flex items, it aligns flex lines.' Option A (align-items) is incorrect because it vertically aligns individual flex items on the cross-axis. Option B (justify-content) is incorrect because it horizontally aligns flex items on the main-axis. Option D (flex-flow) is incorrect because it is a shorthand property for flex-direction and flex-wrap.

Q24 **A mobile device has a physical resolution of 960 x 640 and reports a device pixel ratio (DPR) of 2.0. What is its logical resolution?** HARD

A. 1920 x 1280
B. 640 x 320
C. 960 x 640
D. **480 x 320** **[Correct]**

Correct Answer: D - Explanation:
The Device Pixel Ratio (DPR) is calculated as Physical Resolution divided by Logical Resolution. To find the logical resolution, we divide the physical dimensions by the DPR:
Step 1: 960 / 2 = 480 (logical width)
Step 2: 640 / 2 = 320 (logical height).
This matches the iPhone 4/4S example in the slides where a physical resolution of 960x640 with a DPR of 2 yields a logical resolution of 480x320. Option A is incorrect because it multiplies by the DPR, while Options B and C represent incorrect mathematical divisions.

Q25 Based on the source material, which of the following devices is correctly matched with its reported device pixel ratio? MEDIUM

- A **Nokia Lumia 1020 reports 1.6667** [Correct]
- B Apple iPhone 6 Plus reports 2.00
- C Samsung Galaxy S4 reports 2.46
- D iPhone 4S reports 3.00

Correct Answer: A - Explanation:
According to the slides, the Nokia Lumia 1020 reports a device pixel ratio of 1.6667, making Option A correct. Option B is incorrect because the iPhone 6 Plus reports a ratio of 2.46, not 2.00. Option C is incorrect because the Samsung Galaxy S4 reports a ratio of 3, not 2.46. Option D is incorrect because the iPhone 4S reports a ratio of 2, not 3.00.

Q26 Which meta tag attribute value is critical for defaulting a website to a full 100% zoom on mobile devices? EASY

- A user-scalable=0
- B **initial-scale=1.0** [Correct]
- C maximum-scale=1.0
- D width=device-width

Correct Answer: B - Explanation:
The source states that 'The initial-scale value is important as this defaults your website to a full 100% zoom.' Option A (user-scalable=0) is incorrect because it disables the user's ability to zoom in or out. Option C (maximum-scale=1.0) is incorrect because it limits the maximum zoom level. Option D (width=device-width) is incorrect because it sets the layout viewport width to match the device's screen width.

Q27 What is a key difference between setting 'width: 100%' and 'max-width: 100%' on an image element in CSS? MEDIUM

- A width: 100% prevents the image from scaling up larger than its original size, while max-width: 100% allows it.
- B width: 100% is not supported by Internet Explorer, whereas max-width: 100% is fully supported.
- C **max-width: 100% ensures the image scales down if needed but never scales up past its original size,** [Correct]
- D max-width: 100% causes image distortion when the browser is resized, while width: 100% maintains the aspect ratio.

Correct Answer: C - Explanation:
The source states that if max-width is set to 100%, the image 'will scale down if it has to, but never scale up to be larger than its original size.' Conversely, if width is set to 100%, the image 'will be responsive and scale up and down.' Option A is incorrect because it reverses these definitions. Option B is incorrect because the source notes that Internet Explorer cannot understand max-width: 100% and requires an IE-specific stylesheet with width: 100%. Option D is incorrect because neither property causes distortion if height is set to auto.

Q28 When comparing SVG and Canvas based on the source material, which of the following statements is true? HARD

- A Canvas is resolution independent, while SVG is resolution dependent.
- B Canvas supports event handlers, while SVG has no support for event handlers.
- C Canvas is best suited for Google Maps, while SVG is best suited for game applications.
- **D Canvas allows saving the resulting image as .png or .jpg, while SVG does not mention this capabili [Correct]**

Correct Answer: D - Explanation:
According to the comparison table in the slides, Canvas allows saving the resulting image as .png or .jpg, which makes Option D correct. Option A is incorrect because Canvas is resolution dependent and SVG is resolution independent. Option B is incorrect because Canvas has no support for event handlers, whereas SVG does support them. Option C is incorrect because SVG is best suited for large rendering areas like Google Maps, while Canvas is well suited for game applications.

Q29 Which of the following is NOT listed in the source as an available request header in the current list of HTTP Client Hints? MEDIUM

- **A Content-DPR [Correct]**
- B DPR
- C Viewport-Width
- D Downlink

Correct Answer: A - Explanation:
The source states that the current list of available hints includes 'DPR (device pixel ratio), Width (resource width), Viewport-Width (layout viewport width), and Downlink (maximum downlink speed) request headers.' It specifies that 'Content-DPR' is a response header used to confirm the DPR of selected image resources, not a request header. Therefore, Option A is the correct answer because it is a response header. Options B, C, and D are all valid request headers.

Q30 Which problem of responsive images is described as involving .htaccess and PHP 5.x, which assumes the website is running on an Apache server? HARD

- A Problem 1 - Semantics
- B Problem 4 - Bandwidth Capabilities
- **C Problem 5 - Server Side Components [Correct]**
- D Problem 6 - JavaScript

Correct Answer: C - Explanation:
The source states under 'Problem 5 - Server Side Components' that 'The bulk of Adaptive Images' work is done through .htaccess and PHP 5.x... The problem with .htaccess is that it assumes your website is running on an Apache server.' Option A is incorrect because Semantics deals with downloading unnecessary data. Option B is incorrect because Bandwidth Capabilities deals with adjusting image size based on connection speed. Option D is incorrect because JavaScript deals with avoiding scripts by using third-party solutions.

Q31 **By default, how does the 'background-image' property behave when applied to an HTML element?** EASY

A It displays only once in the top-left corner.
B **It repeats both horizontally and vertically to cover the entire element.** **[Correct]**
C It automatically scales to 100% width and height of the element.
D It centers itself within the element without repeating.

Correct Answer: B - Explanation:
The source states: 'By default, the image is repeated so it covers the entire element.' It also notes that the background-image property repeats an image both horizontally and vertically by default. Option B is therefore correct. Options A, C, and D are incorrect because they describe non-default behaviors that require additional properties like background-repeat: no-repeat or background-size.

Q32 **Which of the following statements is true regarding the CSS 'image-set()' function according to the source?** MEDIUM

A It is fully supported by all modern browsers and ready for live production.
B It requires the '-moz-image-set()' prefix to work in Firefox.
C It is declared within the 'background-repeat' property.
D **It is experimental and only supported in Safari 6 and Google Chrome 21 with the '-webkit-' prefix.** **[Correct]**

Correct Answer: D - Explanation:
The source states: 'CSS Image Set is experimental. It is only supported in Safari 6 and Google Chrome 21 and is prefixed - -webkit-image-set().' Option D is therefore correct. Option A is incorrect because the source explicitly states that with current limited browser support, it is not ready to be implemented in a live site. Option B is incorrect because the Firefox prefix is not mentioned in the source. Option C is incorrect because the function is declared within the background-image property, not background-repeat.

Q33 **According to the studies cited in the source, what is the impact of introducing a performance issue that causes an extra 500 milliseconds (half second) delay in page load?** EASY

A **A slightly more than one percent drop in user engagement and income.** **[Correct]**
B A complete crash of mobile browsers due to memory limits.
C A ten percent drop in search engine ranking.
D No measurable difference since 500ms is too small for users to notice.

Correct Answer: A - Explanation:
The source states: 'Even such a small change [500 milliseconds], causes a slightly more than one percent drop in user engagement, and income from your site.' Option A is therefore correct. Option B is incorrect because a 500ms delay does not cause browser crashes. Option C is incorrect because search engine rankings are not mentioned in this context. Option D is incorrect because users do notice and react to this delay, leading to a drop in engagement.

Q34 What are the three pillars of performance identified in the source material? EASY

A HTML, CSS, JavaScript

B Bandwidth, Latency, Server Speed

C **Network, Render, Compute** [Correct]

D Touch, Mouse, Pointer

Correct Answer: C - Explanation:
The source explicitly lists: 'Three pillars of performance are Network, Render, Compute.' Option C is therefore correct. Option A is incorrect because HTML, CSS, and JavaScript are web technologies, not performance pillars. Option B is incorrect because bandwidth and latency are network-specific metrics. Option D is incorrect because touch, mouse, and pointer are input types.

Q35 What is the minimum recommended size for a touch area to ensure a good user experience on mobile devices? MEDIUM

A 5mm x 5mm

B **10mm x 10mm** [Correct]

C 15mm x 15mm

D 20mm x 20mm

Correct Answer: B - Explanation:
Under 'Simple rules for touch UI', Rule 2 states: 'Size of Touch Area should be at-least 10mm X 10 mm.' Option B is therefore correct. Options A, C, and D are incorrect because they do not match the minimum size specified in the source material.

Q36 Why do mobile browsers typically introduce a click delay of about a third of a second (approx. 300ms)? HARD

A To allow the server to process proactive content negotiation.

B To wait and see if the user is attempting a swipe gesture.

C **To determine if the user is double-tapping to zoom into an element.** [Correct]

D To clear the touch event buffer and prevent accidental double clicks.

Correct Answer: C - Explanation:
The source states: 'Touch supporting mobile browsers, usually interpret double tapping, as a zoom into this element gesture. Unfortunately, this means the platform, delays firing click of it until it can decide whether, the user is double tapping. This delay is about a third of a second.' Option C is therefore correct. Option A is incorrect because content negotiation is a server-side process. Option B is incorrect because swipe gestures are handled differently. Option D is incorrect because it is not meant to prevent accidental double clicks.

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Q37 How do Touch Events differ from Mouse Events regarding event delivery and DOM boundaries? HARD

A Touch events are always delivered to the element that first received the touch and do not walk across [Correct]
B Touch events bubble up through the DOM tree to the document root, whereas mouse events do not.
C Touch events are delivered to whatever element is currently under the finger during movement.
D Touch events automatically emulate mouse clicks only when the touch ends on a different element.

Correct Answer: A - Explanation:
The source states: 'Touch events are always delivered to the element that first received that Touch. They don't walk across boundaries like Mouse Events do.' Option A is therefore correct. Option B is incorrect because touch events do not walk across boundaries. Option C is incorrect because touchmove events continue to target the original element even if the finger moves off it. Option D is incorrect because touch events emulate mouse clicks based on double-tap zoom detection, not boundary crossing.

Q38 Which company made the standards proposal for Pointer Events to reconcile touch, mouse, and pen inputs into a single model? MEDIUM

A Google
B Apple
C W3C
D Microsoft [Correct]

Correct Answer: D - Explanation:
The source states: 'To rectify this situation, Microsoft made a standards proposal that reconciles touch events and mouse events and pen input too, into one model called pointer events.' Option D is therefore correct. Options A, B, and C are incorrect because they did not propose the Pointer Events model according to the text.

Q39 Which Pointer Event attribute is defined as the 'angle of tilt of stylus between Z and X axes, where X and Y plane is on the surface of the screen'? HARD

A tiltY
B tiltX [Correct]
C pressure
D pointerId

Correct Answer: B - Explanation:
According to the pointer attributes table, tiltX is the 'angle of tilt of stylus between Z and X axes, where X and Y plane is on the surface of the screen.' Option B is therefore correct. Option A is incorrect because tiltY is the angle of tilt between Z and Y axes. Option C is incorrect because pressure measures the force of contact. Option D is incorrect because pointerId is a unique numeric identifier.

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Q40 **Which HTML5 input element attribute takes a regular expression (using JavaScript regex syntax) to perform client-side validation?** MEDIUM

- A required
- B placeholder
- C **pattern** **[Correct]**
- D validate

Correct Answer: C - Explanation:
The source states: 'There's also a pattern attribute on the input element, that you can use to validate on the client side. The pattern element takes a regular expression, as expressed by the java script regex syntax.' Option C is therefore correct. Option A is incorrect because required simply specifies that a field must be filled out before submission. Option B is incorrect because placeholder is used for hint text. Option D is incorrect because validate is not a standard HTML5 attribute mentioned in the text.

Q41 **According to the HTML specification, which of the following sets of input types fully supports the 'placeholder' attribute?** EASY

- A **text, search, url, tel, email, password** **[Correct]**
- B text, number, date, file, checkbox, radio
- C text, range, color, submit, button, reset
- D text, search, file, textarea, select, option

Correct Answer: A - Explanation:
The source text explicitly states that the placeholder attribute works with the following input types: text, search, url, tel, email, and password. Option A contains exactly this set of input types, making it the correct answer. Option B is incorrect because input types like checkbox, radio, and file do not support placeholders. Option C is incorrect because range, color, and button elements do not use placeholders. Option D is incorrect because select and option elements do not support the placeholder attribute.

Q42 **If a mobile web developer uses a newly introduced HTML5 input type on an older browser that does not support it, how will the browser handle this input element?** MEDIUM

- A The browser will ignore the element completely and not render anything.
- B The browser will throw a runtime JavaScript error and halt page rendering.
- C **The browser will treat the unrecognized input type as a standard text input.** **[Correct]**
- D The browser will automatically download a polyfill to render the control.

Correct Answer: C - Explanation:
According to the HTML specification, unrecognized input types are to be treated as text by the browser. This ensures backward compatibility, meaning that new input types will still function as text inputs even if they are not supported on older browsers. Option C is correct because it describes this fallback behavior. Option A is incorrect because the browser does not ignore the element. Option B is incorrect because no JavaScript error is thrown. Option D is incorrect because browsers do not automatically download polyfills for unrecognized types.

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Q43 **Why does the HTML best practice recommend wrapping the <label> element around both the <input> element and its associated labeling text?** EASY

A. It is required by the HTML5 specification to make the form submit successfully.
B. **It increases the touchable area of the control, making it much easier to activate on mobile devices[Correct]**
C. It automatically applies a default CSS flexbox layout to align the input and text.
D. It prevents older browsers from rendering the input type as a plain text field.

Correct Answer: B - Explanation:
The source material states that wrapping label elements around input elements and their associated labeling texts increases the touchable area of the control. This makes it much easier for users to activate controls like checkboxes on mobile screens, where precise tapping can be difficult. Therefore, option B is correct. Option A is incorrect because wrapping inputs in labels is a usability best practice, not a strict requirement for form submission. Option C is incorrect because labels do not automatically apply CSS flexbox layouts. Option D is incorrect because labels do not affect how older browsers render unsupported input types.

Q44 **Which of the following statements is TRUE regarding the use of a 'telecolon URL' on mobile and desktop platforms?** EASY

A. It is used to launch the device's default email client with a pre-filled subject line.
B. **It creates a URL that dials a telephone number on mobile, but typically fails on desktop.** **[Correct]**
C. It allows the browser to access the device's GPS coordinates without user permission.
D. It is a specialized CSS selector used to style telephone input fields on mobile screens.

Correct Answer: B - Explanation:
The source text explains that on mobile, a telecolon URL (such as 'tel:') can be used to create a link that dials a telephone number when activated. It also notes that these URLs typically fail on desktop platforms, which is why developers may want to deactivate them there. Therefore, option B is correct. Option A is incorrect because email links use the 'mailto:' scheme, not a telecolon URL. Option C is incorrect because telecolon URLs have no relation to GPS coordinates. Option D is incorrect because it is an HTML URL scheme, not a CSS selector.

Q45 **Based on the source material, which of the following is NOT explicitly listed as a sensor typically found on a mobile device that enables engaging user experiences?** EASY

A. GPS
B. Compass
C. Tilt sensing accelerometer
D. **Barometric pressure sensor** **[Correct]**

Correct Answer: D - Explanation:
The source material lists the sensors typically found on a mobile device as a camera (for stills and video), audio input/output, GPS, a compass, and a tilt-sensing accelerometer. A barometric pressure sensor is not mentioned anywhere in the provided text, making option D the correct answer for this negative question. Options A, B, and C are incorrect because they are explicitly listed in the source text as sensors that enable engaging and interactive user experiences. This highlights how mobile platforms play host to richer experiences compared to traditional desktops.

Q46 **A developer wants to create an HTML file input element that specifically opens the system camera on a mobile device to capture an image. What is the correct syntax for this element?** MEDIUM

- A **<input type="file" accept="image/*;capture=camera">** **[Correct]**
- B <input type="camera" accept="image/*">
- C <input type="file" capture="image/*;camera">
- D <input type="file" accept="camera/*;capture=image">

Correct Answer: A - Explanation:
The source text explicitly provides the correct HTML syntax for accessing the camera on mobile devices as <input type="file" accept="image/*;capture=camera">. This configuration tells the mobile browser to accept images and specifically launch the system camera application, making option A correct. Option B is incorrect because 'camera' is not a valid HTML input type. Option C is incorrect because the 'capture' attribute is not used as a standalone attribute with that syntax in the source. Option D is incorrect because 'camcorder' is used for video, and the attributes are improperly structured.

Q47 **Which HTML code snippet correctly configures a file input element to launch the system camcorder application on a mobile device?** MEDIUM

- A <input type="video" accept="video/*;capture=camcorder">
- B **<input type="file" accept="video/*;capture=camcorder">** **[Correct]**
- C <input type="file" capture="video/*;camcorder">
- D <input type="file" accept="camcorder/*;capture=video">

Correct Answer: B - Explanation:
The source material specifies that the correct syntax for launching the system camcorder on a mobile device is <input type="file" accept="video/*;capture=camcorder">. This makes option B the correct answer. Option A is incorrect because 'video' is not a valid value for the 'type' attribute of an input element. Option C is incorrect because the capture parameter must be part of the accept attribute string as shown in the source. Option D is incorrect because 'camcorder/*' is not a valid MIME type prefix.

Q48 **To allow a mobile user to record audio directly using their device's system microphone, which of the following HTML elements should be implemented?** MEDIUM

- A <input type="audio" accept="audio/*;capture=microphone">
- B <input type="file" capture="audio/*;microphone">
- C **<input type="file" accept="audio/*;capture=microphone">** **[Correct]**
- D <input type="microphone" accept="audio/*">

Correct Answer: C - Explanation:
According to the source text, the correct syntax to access the system microphone for audio recording on mobile is <input type="file" accept="audio/*;capture=microphone">. This makes option C the correct choice. Option A is incorrect because 'audio' is not a valid HTML input type. Option B is incorrect because the capture parameter is misplaced outside the accept attribute. Option D is incorrect because 'microphone' is not a valid input type, and the accept attribute is missing the capture extension.

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Q49 **Which of the following is NOT described in the source material as a drawback of using the HTML capture extension?** HARD

- A It only works on mobile devices; on desktop, it falls back to a normal file open dialogue.
- B The file open button interface is difficult to style and customize.
- C **It requires the installation of third-party plugins on the mobile device to function.** **[Correct]**
- D Redirecting the user to an external system application disrupts the consistent flow of the app.

Correct Answer: C - Explanation:
The source material lists three specific drawbacks of the capture extension: it only works on mobile (not desktop), the file open button is hard to style, and redirecting to an external app disrupts the application's flow. It does not state that third-party plugins are required to make it function, making option C the correct answer. Option A is a documented drawback because desktop users only get a standard file dialogue. Option B is a documented drawback because styling file inputs is notoriously difficult. Option D is a documented drawback because switching apps breaks user experience consistency.

Q50 **Which JavaScript API is used to obtain live, real-time audio and video input directly within a web application without redirecting to an external app?** EASY

- A **navigator.getUserMedia** **[Correct]**
- B navigator.getLiveStream
- C window.captureMedia
- D document.getMediaStream

Correct Answer: A - Explanation:
The source text states that developers can get live input, including audio and video, directly under their application by using an API called getUserMedia. The correct JavaScript interface is navigator.getUserMedia, making option A the correct answer. Option B is incorrect because navigator.getLiveStream is a fabricated API name not mentioned in the text. Option C is incorrect because window.captureMedia does not exist. Option D is incorrect because document.getMediaStream is also a fabricated method.

Q51 **What is the correct method signature and parameter order for the navigator.getUserMedia API as described in the source material?** MEDIUM

- A navigator.getUserMedia(successCallback, errorCallback, constraints);
- B **navigator.getUserMedia(constraints, successCallback, errorCallback);** **[Correct]**
- C navigator.getUserMedia(constraints, errorCallback, successCallback);
- D navigator.getUserMedia(successCallback, constraints);

Correct Answer: B - Explanation:
The source material explicitly defines the syntax for the getUserMedia API as navigator.getUserMedia(constraints, successCallback, errorCallback);. This makes option B the correct answer. Option A is incorrect because the constraints object must be the first parameter, not the last. Option C is incorrect because the success callback must precede the error callback. Option D is incorrect because it completely omits the mandatory error callback parameter.

Q52 When configuring the constraints parameter for navigator.getUserMedia(), which keywords should a developer use to require a specific capability (such as a minimum or exact resolution)? HARD

A require, limit, or force
B **min, max, or exact** **[Correct]**
C low, high, or strict
D atLeast, atMost, or equal

Correct Answer: B - Explanation:
The source text states that to require a specific capability within the constraints object, developers must use the keywords 'min', 'max', or 'exact'. This makes option B the correct answer. Option A is incorrect because 'require', 'limit', and 'force' are not valid constraint keywords. Option C is incorrect because 'low', 'high', and 'strict' are not used for this purpose. Option D is incorrect because 'atLeast', 'atMost', and 'equal' are not part of the MediaStreamConstraints specification.

Q53 To implement a camera application that takes snapshots from a live video stream, what combination of HTML5 elements and methods is recommended by the source material? HARD

A Copying the video stream directly into an <img> tag using video.toDataURL().
B **Drawing the video frame onto a canvas context using drawImage, then encoding it using** **[Correct]**
C Saving the raw binary stream directly to local storage using navigator.saveStream().
D Using a CSS filter to freeze the video element and converting the CSS to an image.

Correct Answer: B - Explanation:
The source material explains that to implement a camera application, developers can grab snapshots from the live video stream by drawing the video frame onto a canvas context using drawImage. Then, because a canvas cannot be directly downloaded or shared, they must encode the canvas contents as an image using the canvas.toDataURL() method. This makes option B the correct answer. Option A is incorrect because video elements do not have a toDataURL() method. Option C is incorrect because there is no navigator.saveStream() method. Option D is incorrect because CSS filters cannot convert video streams into downloadable image files.

Q54 Why is the toDataURL() method of a canvas element necessary when implementing user flows like downloading or sharing snapshots? HARD

A **Because a canvas element cannot be directly copied, downloaded, or posted to social networks without** **[Correct]**
B Because it compresses the video stream to save bandwidth before sending it to the server.
C Because it bypasses browser security restrictions that prevent canvas rendering.
D Because it converts the canvas into a vector SVG format that is scalable.

Correct Answer: A - Explanation:
The source text explains that a canvas element is fundamentally different from an image because users cannot directly copy, download, or post a canvas to social networks. To enable these common user flows, the canvas must be encoded as an image using the toDataURL() method, which copies the canvas contents as a data URL to the source of an image tag. This makes option A the correct answer. Option B is incorrect because the method is not used for network compression of video streams. Option C is incorrect because it does not bypass security restrictions. Option D is incorrect because toDataURL() typically outputs raster formats (like PNG or JPEG), not vector SVGs.

Q55 **In the navigator.getUserMedia() API, what object is passed as the sole argument to the successCallback function when the media stream is successfully acquired?** EASY

A. A MediaStreamError object
B. A CanvasRenderingContext2D object
C. **A MediaStream object** [Correct]
D. An HTMLVideoElement object

Correct Answer: C - Explanation:
According to the source text, when the call to navigator.getUserMedia() succeeds, the function specified in the successCallback is invoked with the MediaStream object. This object contains the active media stream which can then be assigned to a video element. Therefore, option C is correct. Option A is incorrect because MediaStreamError is only passed to the error callback. Option B is incorrect because canvas contexts are not returned by this API. Option D is incorrect because the API returns the stream itself, not an HTML video element.

Q56 **When the call to navigator.getUserMedia() fails, what object is passed as the sole argument to the errorCallback function?** EASY

A. **A MediaStreamError object** [Correct]
B. A PermissionDeniedError string
C. A NavigatorError object
D. A MediaStream object

Correct Answer: A - Explanation:
The source material states that when the call to navigator.getUserMedia() fails, the function specified in the errorCallback is invoked with a MediaStreamError object as its sole argument. This makes option A the correct answer. Option B is incorrect because the callback receives an error object, not a plain string. Option C is incorrect because NavigatorError is not the object specified in the text. Option D is incorrect because a MediaStream object is only returned upon a successful call.

Q57 **Under what condition is the PermissionDeniedError returned to the errorCallback of navigator.getUserMedia()?** EASY

A. When the device does not have a camera or microphone installed.
B. **When permission to use a media device was denied by the user or the system.** [Correct]
C. When the requested resolution constraints cannot be satisfied by the hardware.
D. When the application is running on a desktop browser instead of a mobile browser.

Correct Answer: B - Explanation:
The table in the source material defines PermissionDeniedError as the error that occurs when 'Permission to use a media device was denied by the user or the system.' This makes option B the correct answer. Option A is incorrect because a missing device would trigger a NotFoundError. Option C is incorrect because unsatisfied resolution constraints also trigger a NotFoundError (no tracks satisfy the constraints). Option D is incorrect because running on a desktop browser does not inherently trigger a permission denial error.

Q58 **Which of the following best describes the cause of a NotFoundError in the context of navigator.getUserMedia()?** MEDIUM

- A The user has deleted the browser's cache and local storage.
- B The URL of the web application could not be resolved by the DNS server.
- **C No media tracks of the type specified were found that satisfy the constraints specified.** **[Correct]**
- D The canvas element used to draw the snapshot was not found in the DOM.

Correct Answer: C - Explanation:
The table in the source material defines NotFoundError as the error that occurs when 'No media tracks of the type specified were found that satisfy the constraints specified.' This makes option C the correct answer. Option A is incorrect because clearing browser cache does not cause a media track search failure. Option B is incorrect because DNS resolution is unrelated to local media device discovery. Option D is incorrect because the DOM status of canvas elements does not affect the media stream acquisition process.

Q59 **If a developer specifies <input type="file" accept="image/*">, what does this configuration indicate to the browser?** EASY

- **A The browser should only accept image files.** **[Correct]**
- B The browser should automatically launch the camera and take a picture.
- C The browser should accept any file type but display them as images.
- D The browser should reject all files except those with the .gif extension.

Correct Answer: A - Explanation:
The source text states that <input type="file" accept="image/*"> restricts the file input to 'images only'. This tells the browser to filter the file picker to only allow the selection of image files, making option A the correct answer. Option B is incorrect because this syntax does not include the capture extension required to launch the camera. Option C is incorrect because the accept attribute filters file selection, it does not change how files are rendered. Option D is incorrect because the asterisk wildcard accepts all image formats, not just GIFs.

Q60 **Suppose a developer wants to configure the constraints for navigator.getUserMedia() to demand a minimum resolution of 1280x720. Based on the source material, which keywords or structure should be used within the constraints object?** HARD

- **A width: { min: 1280 }, height: { min: 720 }** **[Correct]**
- B width: { require: 1280 }, height: { require: 720 }
- C resolution: "1280x720"
- D video: { width: 1280, height: 720 }

Correct Answer: A - Explanation:
The source text explains that to require a capability, developers must use keywords like 'min', 'max', or 'exact' within the constraints object, and gives the example of demanding a minimum resolution of 1280x720. This is achieved by nesting these keywords within the width and height properties of the video constraint (e.g., width: { min: 1280 }, height: { min: 720 }), making option A correct. Option B is incorrect because 'require' is not a valid keyword. Option C is incorrect because constraints are structured objects, not plain strings. Option D is incorrect because it specifies exact values without using the required 'min' keyword mentioned in the text.

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Q61 What is the default window width that mobile browsers historically simulated to render desktop websites on small screens? EASY

- A 320 pixels
- B 640 pixels
- C **980 pixels** [Correct]
- D 1024 pixels

Correct Answer: C - Explanation:
According to the source material, when mobile browsers first came along, they 'lied' about the window width to fit desktop content. They made the window act as if it were 980 pixels wide, even though the original iPhone was only 320 pixels across. This allowed sites designed for 1024x768 screens to fit on mobile screens. Option A is the physical width of the original iPhone, Option B is the physical width of later iPhones, and Option D is a standard desktop screen width, making Option C the correct answer.

Q62 A developer designs a fixed-width mobile application that is exactly 916 pixels wide. If the default mobile browser viewport width is 980 pixels, what percentage of the screen width will be wasted (left as blank/underflow space) if no viewport meta tag is set? HARD

- A **6.53%** [Correct]
- B 12.24%
- C 5.00%
- D 8.33%

Correct Answer: A - Explanation:
Formula: Wasted Space Percentage = ((Default Viewport Width - App Width) / Default Viewport Width) * 100
Variables:
Default Viewport Width (D) = 980 pixels
App Width (A) = 916 pixels
Step 1: Calculate the difference in pixels:
Wasted Pixels = D - A
Wasted Pixels = 980 - 916 = 64 pixels
Step 2: Calculate the percentage of wasted space relative to the default width:
Percentage = (Wasted Pixels / D) * 100
Percentage = (64 / 980) * 100
Percentage = 0.0653 * 100 = 6.53%
Therefore, Option A is correct. Options B, C, and D are incorrect calculations that do not use the correct formula or values.

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Q63 To enable Developer Mode on many Android devices, how many times must a user click on the 'Build Number' in the settings menu? EASY

- A 3 times
- B 5 times
- C **7 times** [Correct]
- D 10 times

Correct Answer: C - Explanation:
According to the 'Setup For Mobile' section of the source material, to turn on Developer Mode, you need to go to your device's settings, click on 'About Device', and then click on 'Build Number' seven times. The source explicitly adds 'Seriously' to emphasize this exact number. Options A, B, and D are incorrect because they do not match the specific sequence of seven clicks required by the Android operating system as described in the text.

Q64 **A developer is using CSS Viewport Units to design a square button. If the viewport width is 375 pixels and the viewport height is 812 pixels, and the button's width is set to 20vmin, what is the calculated width of the button in pixels?**

HARD

A 162.4 pixels

B **75.0 pixels** **[Correct]**

C 118.7 pixels

D 37.5 pixels

Correct Answer: B - Explanation:
Formula: vmin_value = (vmin_percentage / 100) * min(Viewport_Width, Viewport_Height)
Variables:
Viewport_Width (W) = 375 pixels
Viewport_Height (H) = 812 pixels
vmin_percentage (P) = 20
Step 1: Identify the smaller dimension between width and height:
min(W, H) = min(375, 812) = 375 pixels
Step 2: Calculate 1% of the smaller dimension:
1vmin = 375 / 100 = 3.75 pixels
Step 3: Multiply by the given percentage (20):
20vmin = 20 * 3.75 = 75.0 pixels
Therefore, Option B is correct. Option A uses the larger dimension (812px * 20% = 162.4px), which represents vmax. Options C and D are incorrect calculations.

Q65 **If a developer sets the viewport width to a fixed 1832 pixels instead of the previous 916 pixels, by what factor does the mobile browser scale down the content to fit the same physical screen?**

MEDIUM

A Factor of 0.5

B Factor of 1.5

C **Factor of 2.0** **[Correct]**

D Factor of 4.0

Correct Answer: C - Explanation:
Formula: Scale Factor = New Viewport Width / Old Viewport Width
Variables:
New Viewport Width (W_new) = 1832 pixels
Old Viewport Width (W_old) = 916 pixels
Step 1: Set up the ratio of the new width to the old width:
Ratio = W_new / W_old
Step 2: Substitute the values:
Ratio = 1832 / 916
Step 3: Solve the division:
Ratio = 2.0
This means the layout is twice as wide, so the browser must scale it down by a factor of 2.0 to fit the same physical screen. Option A is the inverse scale factor (0.5). Options B and D are mathematically incorrect.

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Q66 **Which of the following correctly describes the difference between 'Fluid' and 'Adaptive' web design as defined in the source?** MEDIUM

A Fluid designs use fixed pixel widths, while Adaptive designs use percentages.

B **Fluid designs use percentages for widths, while Adaptive designs introduce media queries to target specific** [Correct]

C Fluid designs use media queries, while Adaptive designs use native app wrappers like PhoneGap.

D Fluid designs are built on a fluid grid, while Adaptive designs do not support mobile devices.

Correct Answer: B - Explanation:
The source states that 'Fluid websites are built using percentages for widths' so columns scale fluidly, whereas 'Adaptive websites introduce media queries to target specific device sizes, like smaller monitors, tablets, and mobile.' Option A reverses these definitions. Option C incorrectly associates PhoneGap with Adaptive design. Option D is incorrect because both designs support mobile devices.

Q67 **Who outlined the 'Mobile First' design approach in 2009?** EASY

A **Luke Wroblewski** [Correct]

B Steve Jobs

C Chrome Developer Team

D W3C Consortium

Correct Answer: A - Explanation:
The source explicitly states: 'Mobile first design is an approach outlined in 2009 by Luke Wroblewski.' It is defined as designing for smaller screens first, then adding more features and content for bigger screens. Options B, C, and D are incorrect because they are not credited with outlining this approach in the provided text.

Q68 **When setting up Reverse Port Forwarding, if a developer runs Python's simple HTTP server on Port 9999 on their local machine, which port rule must be configured in Chrome Inspect to access it on the mobile device?** MEDIUM

A Port 80

B Port 8080

C **Port 9999** [Correct]

D Port 3000

Correct Answer: C - Explanation:
The source explains that to verify the local server is working on Port 9999, the developer accesses it on the local machine and then sets up a port forwarding rule for 'port 9999 on localhost'. Option A, B, and D are incorrect because they do not match the active port of the Python HTTP server (9999) specified in the source.

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Q69 **A responsive layout uses CSS viewport units where a button is designed to take up exactly 1/3rd of the largest square that will fit in the screen. If the viewport dimensions are 600px by 900px, what is the size of this button in pixels using the appropriate CSS viewport unit?**

HARD

A 300 pixels

B **200 pixels** **[Correct]**

C 150 pixels

D 100 pixels

Correct Answer: B - Explanation:
Formula: Button Size = (vmin_percentage / 100) * min(Viewport_Width, Viewport_Height)
Variables:
Viewport_Width (W) = 600 pixels
Viewport_Height (H) = 900 pixels
vmin_percentage (P) = 33.33 (representing 1/3rd of the biggest square)
Step 1: Identify the minimum dimension of the viewport:
min(W, H) = min(600, 900) = 600 pixels
Step 2: Calculate 1/3rd of this minimum dimension:
Button Size = 600 * (33.33 / 100)
Button Size = 600 * (1 / 3)
Button Size = 200 pixels
Therefore, Option B is correct. Option A uses the maximum dimension (900px * 1/3 = 300px), which would use vmax and overflow the width. Options C and D are incorrect calculations.

Q70 **What is a major downside of using a fixed-width viewport on mobile devices according to the source?**

MEDIUM

A The website will crash on desktop browsers.

B **Text and UI elements will look extremely small on small screens, forcing users to zoom.** **[Correct]**

C The layout will automatically convert into a native app.

D It disables all CSS styling on mobile devices.

Correct Answer: B - Explanation:
The source states that using a fixed viewport means 'your content is almost always going to be scaled' and 'in some cases, users are going to have to zoom, to be able to see or read anything, particularly text.' Option A is incorrect because desktop browsers do not respond to the viewport tag at all. Options C and D are completely false and not mentioned in the source.

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Q71 **A developer is testing an iOS device. If the portrait width of the device is 320 pixels and the landscape width is 480 pixels, and the viewport meta tag is set to '<meta name="viewport" content="width=device-width">' without setting initial-scale=1, what width does the iOS web engine assume when the device is rotated to landscape?** HARD

A. 480 pixels
B. **320 pixels** [Correct]
C. 980 pixels
D. 640 pixels

Correct Answer: B - Explanation:
Concept: iOS viewport scaling behavior without initial-scale.
Variables:
Portrait Width (W_portrait) = 320 pixels
Landscape Width (W_landscape) = 480 pixels
Step 1: Identify the behavior of iOS when only width=device-width is set. The source states that the iOS web engine keeps the same viewport width and rescales it to fit across the landscape screen.
Step 2: Determine the assumed width:
Assumed Width = W_portrait = 320 pixels
Step 3: Compare with options. Option B is correct because iOS maintains the portrait width of 320 pixels and simply stretches it. Option A (480 pixels) is only recognized if initial-scale=1 is set. Options C and D are incorrect.

Q72 **What is the design philosophy of stripping down to your core user experience and then layering additional features on top if space permits?** EASY

A. Desktop First
B. **Mobile First** [Correct]
C. Fluid Grid Layout
D. Native App Wrapping

Correct Answer: B - Explanation:
The source defines 'Mobile First' as 'really a design philosophy of stripping down to your core user experience and then layering additional gravy on top if you have the appropriate space.' Option A is the opposite. Options C and D are technical layout/wrapper terms, not this specific design philosophy.

Q73 **Which of the following is NOT one of the three simple rules mentioned in the source for offering a mobile-only site?** MEDIUM

A. Make sure a user can still get to the full site from the mobile site.
B. Put canonical URL in meta-information.
C. **Transfer users linking to a specific desktop page to the top-level mobile page.** [Correct]
D. Transfer users linking to a specific desktop page to its equivalent page on the mobile site.

Correct Answer: C - Explanation:
The source explicitly states: 'if someone links to a specific page on your desktop site, you shouldn't transfer them to your top-level mobile page. You need to transfer them to that specific page or its equivalent.' Therefore, transferring them to the top-level page is a violation of the rules (it is frustrating). Options A, B, and D are correct rules mentioned in the text.

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Q74 **Which CSS layout tool, recently shipped in iOS 7 and Android, lets you stack elements in flexible rows or columns?** EASY

A Grid Layout
B Float Layout
C **FlexBox** [Correct]
D Absolute Positioning

Correct Answer: C - Explanation:
The source states: 'One tool that makes reflow a lot easier... is the new flex box layout in CSS. This new tool lets you stack elements in flexible rows or columns.' Options A, B, and D are other CSS layout methods but are not the specific tool highlighted in the text.

Q75 **If you want to prevent Android from incorrectly zooming and rescaling a page when it forcibly overflows during orientation changes, what is the workaround mentioned in the source?** MEDIUM

A Set user-scalable to yes.
B **Set minimum-scale and maximum-scale properties to 1.** [Correct]
C Remove the viewport meta tag entirely.
D Use Python's simple HTTP server.

Correct Answer: B - Explanation:
The source states: 'the only way that I found, to get around this... is to set the minimum scale and maximum scale properties, to one, which will prevent the user from zooming altogether.' Option A is incorrect because disabling zoom requires setting user-scalable to no or setting min/max scale to 1. Option C would revert to the default 980px viewport. Option D is a server tool, not a viewport workaround.

Q76 **A developer is configuring a viewport meta tag. If they want to set a fixed viewport width that is exactly double the default portrait screen width of an iPhone (320 pixels), what value should they assign to the width property?** MEDIUM

A 320
B **640** [Correct]
C 980
D 1280

Correct Answer: B - Explanation:
Formula: Target Width = Base Portrait Width * Scale Factor
Variables:
Base Portrait Width (W_base) = 320 pixels
Scale Factor (S) = 2
Step 1: Identify the base portrait width of the iPhone from the source: 320 pixels.
Step 2: Multiply by the scale factor:
Target Width = 320 * 2
Step 3: Solve the multiplication:
Target Width = 640 pixels.
Therefore, Option B is correct. Option A is the single portrait width. Option C is the default simulated viewport width (980px). Option D is the Nexus 7 landscape width (1280px).

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Q77 What is the primary difference between traditional websites and mobile websites regarding content depth according to the source?

EASY

A. Traditional websites focus on essential information, while mobile websites require depth and completeness.
B. **Traditional websites require depth and completeness, while mobile users only access essential information** [Correct]
C. Both traditional and mobile websites must contain identical layouts and secondary information.
D. Mobile websites must contain more secondary information than traditional websites.

Correct Answer: B - Explanation:
The source states: 'On a traditional website, depth and completeness are king... In many ways this is the opposite to a mobile web page as a mobile user only accesses essential information and functions.' Option A reverses this. Options C and D contradict the source.

Q78 Why does the source advise against blocking access to a web application with a full-page ad for a native mobile app?

MEDIUM

A. Because native apps do not work on mobile browsers.
B. **Because if a user opens a URL in a browser, they probably want to stay in the browser.** [Correct]
C. Because full-page ads disable the viewport meta tag.
D. Because smart app banners are only supported on Android.

Correct Answer: B - Explanation:
The source states: 'you shouldn't block access to your web application with a full page ad for your native mobile app... if I open a URL in a browser, I probably want to stay in the browser.' Option A is false. Option C is technically incorrect and not in the text. Option D is incorrect because the source specifically mentions smart app banners in iOS.

Q79 What is the 'viewport' as defined in the source?

EASY

A. The physical size of the mobile device's casing.
B. **The user's visible area of a web page.** [Correct]
C. A native app wrapper like PhoneGap.
D. The server port used for reverse port forwarding.

Correct Answer: B - Explanation:
The source explicitly defines: 'The viewport is the user's visible area of a web page.' It varies with the device and is smaller on a mobile phone than on a computer screen. Options A, C, and D are incorrect definitions not supported by the text.

Q80 What happens to a desktop browser when you change or configure the viewport meta tag in your HTML?

MEDIUM

A. The desktop browser will scale the page down to 320 pixels.
B. The desktop browser will throw a syntax error.
C. **Non-mobile desktop browsers do not respond to the viewport tag at all.** [Correct]
D. The desktop browser will automatically enable touch navigation.

Correct Answer: C - Explanation:
The source states: 'your text in UI will not change at all on the desktop. Because, currently, at least, none mobile browsers, don't respond to the view port at all.' (Note: 'none mobile' is a typo in source for 'non-mobile'). Options A, B, and D are common misconceptions but incorrect according to the source.

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Q81 **Which CSS rule is used in Media Queries to conditionally include a block of CSS properties only if a certain condition is true?** EASY

- A **@media** [Correct]
- B @import
- C @supports
- D @viewport

Correct Answer: A - Explanation:
According to the source material, Media Queries use the @media rule to include a block of CSS properties only if a certain condition is true. The @import rule is used to import other stylesheets, while @supports is used for feature queries. Therefore, @media is the correct rule specifically tailored for media-dependent stylesheets.

Q82 **According to the source material, which of the following is NOT listed as an available Media Query expression feature?** MEDIUM

- A Device-aspect-ratio
- B Monochrome
- C Color-index
- D **Pixel-density** [Correct]

Correct Answer: D - Explanation:
The source material explicitly lists the available Media Query expressions: Width, Height, Device-width, Device-height, Orientation, Aspect-ratio, Device-aspect-ratio, Color, Color-index, Monochrome, Grid, and Resolution. 'Pixel-density' is not included in this list. Instead, resolution and device pixel ratio are handled through other mechanisms or the 'Resolution' feature.

Q83 **Which of the following is a key benefit of the CSS3 Flexbox model over the traditional block model?** EASY

- A It relies heavily on floats for positioning elements.
- B **It does not use floats, and the flex container's margins do not collapse with the margins of its contents.** [Correct]
- C It automatically converts all raster images to vector graphics.
- D It forces margins to collapse to save layout space.

Correct Answer: B - Explanation:
The source text states that the flexible box model provides an improvement over the block model because it does not use floats. Additionally, the flex container's margins do not collapse with the margins of its contents. This ensures that elements behave predictably across different screen sizes.

Q84 **Which Flexbox property modifies the behavior of the flex-wrap property by aligning flex lines instead of individual flex items?** MEDIUM

- A align-items
- B justify-content
- C **align-content** [Correct]
- D flex-flow

Correct Answer: C - Explanation:
According to the Flexbox Properties table in the source, the 'align-content' property modifies the behavior of the flex-wrap property. It is similar to align-items, but instead of aligning individual flex items, it aligns flex lines. 'align-items' vertically aligns items on the cross-axis, 'justify-content' horizontally aligns items on the main-axis, and 'flex-flow' is a shorthand for flex-direction and flex-wrap.

Q85 **If a mobile device has a physical resolution of 960 x 640 and reports a Device Pixel Ratio (DPR) of 2.0, what is its logical resolution?** HARD

- A 1920 x 1280
- B **480 x 320** [Correct]
- C 960 x 640
- D 240 x 160

Correct Answer: B - Explanation:
The Device Pixel Ratio is defined as the ratio between physical pixels and logical pixels: DPR = Physical Resolution / Logical Resolution.
To find the logical resolution, we rearrange the formula: Logical Resolution = Physical Resolution / DPR.
Step 1: Divide physical width by DPR: 960 / 2.0 = 480.
Step 2: Divide physical height by DPR: 640 / 2.0 = 320.
Therefore, the logical resolution is 480 x 320.

Q86 **Based on the source material, which of the following devices is correctly matched with its reported Device Pixel Ratio (DPR)?** MEDIUM

- A **Nokia Lumia 1020 reports 1.6667** [Correct]
- B Samsung Galaxy S4 reports 2.46
- C Apple iPhone 6 Plus reports 3.0
- D Apple iPhone 4 reports 1.0

Correct Answer: A - Explanation:
The source material explicitly states that the Nokia Lumia 1020 reports a DPR of 1.6667, the Samsung Galaxy S4 reports 3, and the Apple iPhone 6 Plus reports 2.46. The iPhone 4 and iPhone 4S report a DPR of 2. Therefore, Option A is the only correct match.

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Q87 **In the viewport meta tag '<meta name="viewport" content="width=device-width; initial-scale=1.0; maximum-scale=1.0; user-scalable=0;" />', what is the primary purpose of setting 'initial-scale=1.0'?** MEDIUM

A. It disables user zooming completely.
B. **It defaults the website to a full 100% zoom.** **[Correct]**
C. It forces the browser to render in landscape mode.
D. It sets the physical pixel ratio to exactly 1.0.

Correct Answer: B - Explanation:
The source text states that the 'initial-scale' value is important because it defaults your website to a full 100% zoom. Disabling user zooming is handled by 'user-scalable=0' or setting minimum and maximum scales to 1.0. It does not force landscape mode or alter the physical pixel ratio.

Q88 **When implementing responsive images in CSS, what is a key difference between using 'max-width: 100%' and 'width: 100%'?** MEDIUM

A. Setting 'width: 100%' prevents the image from scaling up larger than its original size.
B. **Setting 'max-width: 100%' allows the image to scale down if needed but prevents it from scaling up larger** **[Correct]**
C. Setting 'max-width: 100%' is fully understood by Internet Explorer without any specific stylesheet.
D. Setting 'width: 100%' is only used for vector graphics like SVG.

Correct Answer: B - Explanation:
The source material explains that if the 'max-width' property is set to 100%, the image will scale down if it has to, but never scale up to be larger than its original size. Setting 'width: 100%' causes the image to scale both up and down, which can cause distortion. Internet Explorer cannot understand 'max-width: 100%', so an IE-specific stylesheet using 'width: 100%' is required.

Q89 **Which of the following is a characteristic of HTML Canvas when compared to SVG?** MEDIUM

A. It is resolution independent.
B. It supports built-in event handlers.
C. **It is well suited for graphic-intensive games.** **[Correct]**
D. It is best suited for applications with large rendering areas like Google Maps.

Correct Answer: C - Explanation:
According to the comparison table in the source, Canvas is resolution dependent, has no support for event handlers, and is well suited for graphic-intensive games. SVG is resolution independent, supports event handlers, and is best suited for large rendering areas like Google Maps. Therefore, Option C is the correct characteristic of Canvas.

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Q90 **Which HTTP request header is used by the client to indicate its device pixel ratio, and which response header is used by the server to confirm the DPR of the selected image resource?** HARD

A Request: CH-DPR; Response: Content-DPR
B **Request: DPR; Response: Content-DPR** **[Correct]**
C Request: Viewport-Width; Response: CH-RW
D Request: DPR; Response: CH-DPR

Correct Answer: B - Explanation:
The source text states that the client sends the 'DPR' request header (e.g., 'DPR: 2.0' in the GET request example) to indicate its device pixel ratio. The server then uses the 'Content-DPR' response header to confirm the DPR of the selected image resources. Option B correctly identifies these headers as described in the HTTP Client Hints specification.

Q91 **Which of the following describes the 'Art Direction' problem in responsive images as defined in the source material?** MEDIUM

A Downloading unnecessary data when the source points to a real image with alt text.
B **The message of an image being diluted or lost when providing several resolution versions of an image.** **[Correct]**
C The layout crashing due to invalid HTML syntax.
D The requirement of running Apache servers to parse .htaccess files.

Correct Answer: B - Explanation:
The source defines 'Problem 2 - Art Direction' as a situation where providing several resolution versions of an image can sometimes have a negative effect where the message of the image is diluted or lost altogether. Option A describes the Semantics problem, Option C describes the Validity problem, and Option D describes the Server Side Components problem.

Q92 **By default, how does the 'background-image' property behave in CSS if the image is smaller than the element it is applied to?** EASY

A It stretches to fit the entire element.
B It is displayed only once in the top-left corner.
C **It is repeated both horizontally and vertically to cover the entire element.** **[Correct]**
D It is automatically centered and cropped.

Correct Answer: C - Explanation:
The source text states that by default, the background-image property repeats an image both horizontally and vertically so it covers the entire element. To prevent this, developers must explicitly set 'background-repeat: no-repeat'.

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Q93 **Why is the CSS 'image-set()' function considered a better approach for high-resolution displays compared to traditional media queries?** HARD

A **It allows the browser discretion to choose the appropriate image based on connection speed and resolution,** [Correct]
B It is fully supported by all legacy browsers without any vendor prefixes.
C It automatically converts raster images into SVG vector graphics on the fly.
D It forces the server to dynamically crop images based on the viewport width.

Correct Answer: A - Explanation:
The source states that media queries don't give the browser any discretion (they explicitly force a source based on density), and they separate image sources by dozens of lines of CSS. 'image-set()' keeps resolutions in the same place and can deliver the most appropriate image based on connection speed. Option B is incorrect because it is experimental and requires '-webkit-image-set()'. Options C and D are incorrect as it does not convert formats or force server-side cropping.

Q94 **According to studies cited in the source material, what is the impact of a 500-millisecond (half-second) delay in page load time?** EASY

A A negligible change that users do not notice.
B **A slightly more than one percent drop in user engagement and income.** [Correct]
C A complete crash of mobile web browsers.
D A ten percent increase in server bandwidth consumption.

Correct Answer: B - Explanation:
The source states that introducing a performance issue causing an extra half second (500 milliseconds) to load causes a 'slightly more than one percent drop in user engagement, and income from your site.' This demonstrates that speed is a requirement, not just a feature.

Q95 **Which of the following is NOT one of the simple rules for touch UI design listed in the source material?** MEDIUM

A Do not rely on Hover Effect.
B Size of Touch Area should be at least 10mm x 10mm.
C **Touch and mouse are mutually exclusive.** [Correct]
D Provide Mouse support as well.

Correct Answer: C - Explanation:
Rule 3 in the source explicitly states 'Touch and mouse are not mutually exclusive.' Therefore, Option C is the incorrect statement (and the correct answer). Rules 1, 2, and 4 correspond to Options A, B, and D respectively.

Q96 **Why do mobile browsers traditionally introduce a click delay of about a third of a second, and how can a developer resolve this without using external libraries?** HARD

A To allow the rendering engine to load high-resolution images; resolved by setting 'image-set()'.
B **To interpret whether the user is performing a double-tap to zoom gesture; resolved by setting the viewport** [Correct]
C To wait for mouse emulation events to complete; resolved by using 'pointerdown' instead of 'mousedown'.
D To clear the touch event buffer; resolved by adding a 'touchcancel' handler.

Correct Answer: B - Explanation:
The source explains that mobile browsers delay firing click events to decide if the user is double-tapping to zoom. This delay is about a third of a second. One way to fix this without libraries is to set the viewport to be non-scalable (e.g., setting 'user-scalable=no' or setting minimum/maximum scale to 1.0).

Q97 **Which of the following is a critical difference in how Touch Events are delivered compared to Mouse Events in the DOM?** MEDIUM

A Touch events walk across DOM boundaries to find the topmost element.
B **Touch events are always delivered to the element that first received the touch and do not walk across** [Correct]
C Touch events are only fired on the document body.
D Touch events automatically trigger mouse emulation on all sibling elements.

Correct Answer: B - Explanation:
The source states: 'Touch events are always delivered to the element that first received that Touch. They don't walk across boundaries like Mouse Events do.' This is why it is important to hook the right elements and why dragging across elements doesn't naturally shift the target of the touch events.

Q98 **Which standard proposal was created by Microsoft to reconcile touch events, mouse events, and pen input into a single unified model?** MEDIUM

A Unified Input API
B **Pointer Events** [Correct]
C Hybrid Touch Model
D Gesture Events

Correct Answer: B - Explanation:
The source material states: 'To rectify this situation, Microsoft made a standards proposal that reconciles touch events and mouse events and pen input too, into one model called pointer events.'

Q99 **Which of the following is an event type defined by the PointerEvent interface that is fired when the active buttons state is entered (e.g., when contact is made with the screen for touch/stylus, or when a button is pressed for a mouse)?** HARD

A pointerover
B pointerenter
C **pointerdown** [Correct]
D gotpointercapture

Correct Answer: C - Explanation:
According to the Event types table in the source, 'pointerdown' is fired when the active buttons state is entered: for touch and stylus, this is when contact is made with the screen; for mouse, when a button is pressed. 'pointerover' is fired when the pointer moves over an element's hit test boundaries, 'pointerenter' is similar but doesn't bubble, and 'gotpointercapture' is fired when an element becomes the target of pointer capture.

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Q100 **Consider the following HTML input element: '<input type="text" pattern="[A-Z]{3}" required placeholder="PAK">'. Which of the following inputs would fail client-side validation when submitting this form?**

MEDIUM

A PAK

B LHR

C **pk** **[Correct]**

D ISB

Correct Answer: C - Explanation:
The 'pattern' attribute uses a regular expression '[A-Z]{3}', which requires exactly three uppercase letters.
- 'PAK', 'LHR', and 'ISB' all consist of exactly three uppercase letters and will pass validation.
- 'pk' consists of only two lowercase letters, failing both the uppercase requirement and the length requirement of exactly three characters. Thus, Option C is the correct answer.

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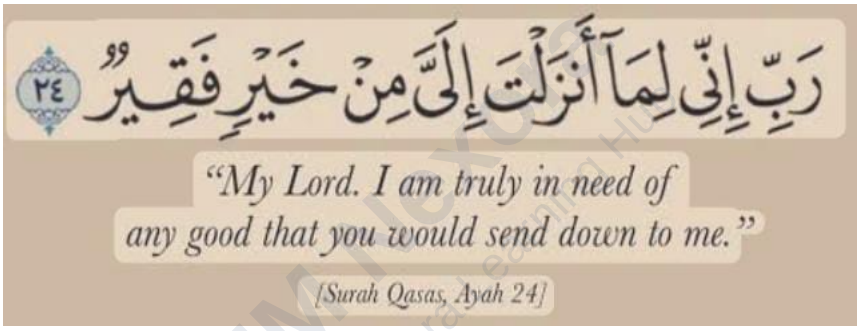

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